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8 April 1971
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# MILITARY STANDARD

# SCREW-THREAD, MODIFIED, 60° STUB, DOUBLE

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MIL-STD-1373
8 April 1971

DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20360

Screw-Thread, Modified,
60° Stub, Double
MIL-STD-1373

1. This standard is mandatory for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions, or deletions should be addressed to Commander, Naval Electronic Systems Command, Department of the Navy, Washington, D.C. 20360.



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# SCREW-THREAD, MODIFIED, 60° STUB, DOUBLE

## 1. SCOPE

1.1 This standard covers the requirements for the double stub, 60° modified screw-thread series, classes and sizes for quick disconnect use.

1.2 Classification. Screw-threads covered by this standard shall be of the following series and classes, as specified.

| | | Double Stub Thread Series | |
|---|---|---|---|
| Class 2A | Symbols DS-2A | .05 pitch | Table I |
| | | 0.1 pitch | Table II |
| | | .1428 pitch | Table III |
| Class 2B | Symbols DS-2B | .05 pitch | Table IV |
| | | 0.1 pitch | Table V |
| | | .1428 pitch | Table VI |

1.3 Designation

| | |
|---|---|
| Class 2A | X.XXXX (size) - 0.05P-0.1L-DS-2A |
| | X.XXXX (size) - 0.1P-0.2L-DS-2A |
| | X.XXXX (size) - .1428P-.2857L-DS-2A |
| Class 2B | X.XXXX (size) - 0.05P-0.1L-DS-2B |
| | X.XXXX (size) - 0.1P-0.2L-DS-2B |
| | X.XXXX (size) - .1428P-.2857L-DS-2B |

## 2. REFERENCED DOCUMENTS

2.1 The issue of the following documents in effect on date of invitation for bids forms a part of this standard to the extent specified herein.

GOVERNMENTAL

STANDARDS

MILITARY

MIL-STD-10 - Surface Roughness, Waviness and Lay

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402)

NONGOVERNMENTAL

Commercial Standard AGD CS8-51 - Gage Blanks.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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TABLE I. Class 2A
.05 Pitch
Modified 60° Stub External Double Thread Series
Symbol DS-2A

| Designation | | | Allowance | External Thread—Limits of Size | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Major Diameter | | | Pitch Diameter | | | Minor Diameter | |
| | | | | Limits | | Tolerance | Limits | | Tolerance | Limits | |
| Thread Size | Pitch | Lead | | Max | Min | | Max | Min | | Max | Min |
| .2500 | .05 | .1 | .0015 | .2485 | .2425 | .0060 | .2305 | .2245 | .0060 | .2105 | .2005 |
| .3750 | ↓ | ↓ | ↓ | .3735 | .3675 | ↓ | .3555 | .3495 | ↓ | .3355 | .3255 |
| .5000 | ↓ | ↓ | ↓ | .4985 | .4925 | ↓ | .4805 | .4745 | ↓ | .4605 | .4505 |
| .5625 | ↓ | ↓ | ↓ | .5610 | .5550 | ↓ | .5430 | .5370 | ↓ | .5230 | .5130 |
| .6250 | ↓ | ↓ | ↓ | .6235 | .6175 | ↓ | .6055 | .5995 | ↓ | .5855 | .5755 |

Note: Formulas for these values are given in table VII.

TABLE II. Class 2A
0.1 Pitch
Modified 60° Stub External Double Thread Series
Symbol DS-2A

| Designation | | | Allowance | External Thread—Limits of Size | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Major Diameter | | | Pitch Diameter | | | Minor Diameter | |
| | | | | Limits | | Tolerance | Limits | | Tolerance | Limits | |
| Thread Size | Pitch | Lead | | Max | Min | | Max | Min | | Max | Min |
| .6875 | .1 | .2 | .0015 | .6860 | .6780 | .0080 | .6600 | .6520 | .0080 | .6300 | .6160 |
| .7500 | ↓ | ↓ | ↓ | .7485 | .7405 | ↓ | .7225 | .7145 | ↓ | .6925 | .6785 |
| .8750 | ↓ | ↓ | ↓ | .8735 | .8655 | ↓ | .8475 | .8395 | ↓ | .8175 | .8035 |
| 1.0000 | ↓ | ↓ | ↓ | .9985 | .9905 | ↓ | .9725 | .9645 | ↓ | .9425 | .9285 |
| 1.1250 | ↓ | ↓ | .0020 | 1.1230 | 1.1110 | .0120 | 1.0890 | 1.0790 | .0100 | 1.0510 | 1.0330 |
| 1.2500 | ↓ | ↓ | ↓ | 1.2480 | 1.2360 | ↓ | 1.2140 | 1.2040 | ↓ | 1.1760 | 1.1580 |
| 1.3750 | ↓ | ↓ | ↓ | 1.3730 | 1.3610 | ↓ | 1.3390 | 1.3290 | ↓ | 1.3010 | 1.2830 |
| 1.5000 | ↓ | ↓ | ↓ | 1.4980 | 1.4860 | ↓ | 1.4640 | 1.4540 | ↓ | 1.4260 | 1.4080 |
| 1.7500 | ↓ | ↓ | ↓ | 1.7480 | 1.7360 | ↓ | 1.7140 | 1.7040 | ↓ | 1.6760 | 1.6580 |
| 2.0000 | ↓ | ↓ | ↓ | 1.9980 | 1.9860 | ↓ | 1.9640 | 1.9540 | ↓ | 1.9260 | 1.9080 |
| 2.2500 | ↓ | ↓ | ↓ | 2.2480 | 2.2360 | ↓ | 2.2140 | 2.2040 | ↓ | 2.1760 | 2.1580 |
| 2.5000 | ↓ | ↓ | ↓ | 2.4980 | 2.4860 | ↓ | 2.4640 | 2.4540 | ↓ | 2.4260 | 2.4080 |
| 2.6250 | ↓ | ↓ | ↓ | 2.6230 | 2.6110 | ↓ | 2.5890 | 2.5790 | ↓ | 2.5510 | 2.5330 |
| 2.7500 | ↓ | ↓ | ↓ | 2.7480 | 2.7360 | ↓ | 2.7140 | 2.7040 | ↓ | 2.6760 | 2.6580 |
| 3.0000 | ↓ | ↓ | ↓ | 2.9980 | 2.9860 | ↓ | 2.9640 | 2.9540 | ↓ | 2.9260 | 2.9080 |
| 3.2500 | ↓ | ↓ | ↓ | 3.2480 | 3.2360 | ↓ | 3.2140 | 3.2040 | ↓ | 3.1760 | 3.1580 |

Note: Formulas for these values are given in table VII.



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TABLE III. Class 2A
.1428 Pitch
Modified 60° Stub External Double Thread Series
Symbol DS-2A

| Designation | | | Allowance | External Thread—Limits of Size | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Major Diameter | | | Pitch Diameter | | | Minor Diameter | | |
| | | | | Limits | | Tolerance | Limits | | Tolerance | Limits | | |
| Thread Size | Pitch | Lead | | Max | Min | | Max | Min | | Max | Min | |
| 2.0000 | .1428 | .2857 | .0020 | 1.9980 | 1.9860 | .0120 | 1.9360 | 1.9260 | .0100 | 1.8685 | 1.8505 | |
| 2.2500 | ↓ | ↓ | ↓ | 2.2480 | 2.2360 | ↓ | 2.1860 | 2.1760 | ↓ | 2.1185 | 2.1005 | |
| 2.7500 | | | | 2.7480 | 2.7360 | | 2.6860 | 2.6760 | | 2.6185 | 2.6005 | |
| 3.0000 | | | | 2.9980 | 2.9860 | | 2.9360 | 2.9260 | | 2.8685 | 2.8505 | |
| 3.2500 | | | | 3.2480 | 3.2360 | | 3.1860 | 3.1760 | | 3.1185 | 3.1005 | |

Note: Formulas for these values are given in table VII.

TABLE IV. Class 2B
.05 Pitch
Modified 60° Stub Internal Double Thread Series
Symbols DS-2B

| Designation | | | Internal Thread—Limits of Size | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Minor Diameter | | | Pitch Diameter | | | Major Diameter | | |
| | | | Limits | | Tolerance | Limits | | Tolerance | Limits | | |
| Thread Size | Pitch | Lead | Min | Max | | Min | Max | | Min | Max | |
| .2500 | .05 | .1 | .2183 | .2263 | .0080 | .2320 | .2400 | .0080 | .2520 | .3640 | |
| .3750 | ↓ | ↓ | .3433 | .3513 | ↓ | .3570 | .3650 | ↓ | .3770 | .3890 | |
| .5000 | | | .4683 | .4763 | | .4820 | .4900 | | .5020 | .5140 | |
| .5625 | | | .5308 | .5388 | | .5445 | .5525 | | .5645 | .5765 | |
| .6250 | | | .5933 | .6013 | | .6070 | .6150 | | .6270 | .6390 | |

Note: Formulas for these values are given in table VII.

TABLE V. Class 2B
0.1 Pitch
Modified 60° Stub Internal Double Thread Series
Symbol DS-2B

| Designation | | | Internal Thread—Limits of Size | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Minor Diameter | | | Pitch Diameter | | | Major Diameter | | |
| | | | Limits | | Tolerance | Limits | | Tolerance | Limits | | |
| Thread Size | Pitch | Lead | Min | Max | | Min | Max | | Min | Max | |
| .6875 | .1 | .2 | .6417 | .6517 | .0100 | .6615 | .6715 | .0100 | .6915 | .7075 | |
| .7500 | ↓ | ↓ | .7042 | .7142 | ↓ | .7240 | .7340 | ↓ | .7540 | .7700 | |
| .8750 | | | .8292 | .8392 | | .8490 | .8590 | | .8790 | .8950 | |
| 1.0000 | | | .9542 | .9642 | | .9740 | .9840 | | 1.0040 | 1.0200 | |



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TABLE V. Class 2B
0.1 Pitch
Modified 60° Stub Internal Double Thread Series
Symbol DS-2B (Continued)

| Designation | | | Internal Thread—Limits of Size | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Minor Diameter | | | Pitch Diameter | | | Major Diameter | |
| Thread Size | Pitch | Lead | Limits | | Tolerance | Limits | | Tolerance | Limits | |
| | | | Min | Max | | Min | Max | | Min | Max |
| 1.1250 | .1 | .2 | 1.0650 | 1.0770 | .0120 | 1.0910 | 1.1030 | .0120 | 1.1290 | 1.1490 |
| 1.2500 | | | 1.1900 | 1.2020 | | 1.2160 | 1.2280 | | 1.2540 | 1.2740 |
| 1.3750 | | | 1.3150 | 1.3270 | | 1.3410 | 1.3530 | | 1.3790 | 1.3990 |
| 1.5000 | | | 1.4400 | 1.4520 | | 1.4660 | 1.4780 | | 1.5040 | 1.5240 |
| 1.7500 | | | 1.6900 | 1.7020 | | 1.7160 | 1.7280 | | 1.7540 | 1.7740 |
| 2.0000 | | | 1.9400 | 1.9520 | | 1.9660 | 1.9780 | | 2.0040 | 2.0240 |
| 2.2500 | | | 2.1900 | 2.2020 | | 2.2160 | 2.2280 | | 2.2540 | 2.2740 |
| 2.5000 | | | 2.4400 | 2.4520 | | 2.4660 | 2.4780 | | 2.5040 | 2.5240 |
| 2.6250 | | | 2.5650 | 2.5770 | | 2.5910 | 2.6030 | | 2.6290 | 2.6490 |
| 2.7500 | | | 2.6900 | 2.7020 | | 2.7160 | 2.7280 | | 2.7540 | 2.7740 |
| 3.0000 | | | 2.9400 | 2.9520 | | 2.9660 | 2.9780 | | 3.0040 | 3.0240 |
| 3.2500 | | | 3.1900 | 3.2020 | | 3.2160 | 3.2280 | | 3.2540 | 3.2740 |

Note: Formulas for these values are given in table VII.

TABLE VI. Class 2B
.1428 Pitch
Modified 60° Stub Internal Double Thread Series
Symbol DS-2B

| Designation | | | Internal Thread—Limits of Size | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Minor Diameter | | | Pitch Diameter | | | Major Diameter | |
| Thread Size | Pitch | Lead | Limits | | Tolerance | Limits | | Tolerance | Limits | |
| | | | Min | Max | | Min | Max | | Min | Max |
| 2.0000 | .1428 | .2857 | 1.8908 | 1.9028 | .0120 | 1.9380 | 1.9500 | .0120 | 2.0055 | 2.0255 |
| 2.2500 | | | 2.1408 | 2.1528 | | 2.1880 | 2.2000 | | 2.2555 | 2.2755 |
| 2.7500 | | | 2.6408 | 2.6528 | | 2.6880 | 2.7000 | | 2.7555 | 2.7755 |
| 3.0000 | | | 2.8908 | 2.9028 | | 2.9380 | 2.9500 | | 3.0055 | 3.0255 |
| 3.2500 | | | 3.1408 | 3.1528 | | 3.1880 | 3.2000 | | 3.2555 | 3.2755 |

Note: Formulas for these values are given in table VII.



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TABLE VII. Thread Data

| Gage | Major Diameter | | | | | Pitch Diameter | | Minor Diameter | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Tolerance Class | Dimension Dg: Full | Dimension Dg: Tolerance | Dimension Dg: Truncated | Dimension Dg: Tolerance | Width of Relief | Dimension Eg | Tolerance | Dimension Kg | Tolerance | Width of Relief |
| **Truncated Setting Plugs** | | | | | | | | | | | |
| Go -2A | Y | max Ds | plus | min Ds | minus | | max Es | minus | | | Rs max |
| No Go -2A | X | max Ds | plus | min Es+2h/3 | minus | | min Es | plus | | | Rs max |
| Go -3A | Y | max Ds | plus | min Ds | minus | | max Es + A | minus | | | Rs max |
| **Thread Ring Gages** | | | | | | | | | | | |
| Go -2A | Y | | | | | Rc max | max Es | minus | min Kn - A | minus | |
| No Go -2A | X | | | | | Ra max | min Es | plus | min Kn + .05h | plus | |
| Go -3A | Y | | | | | Rc max | max Es + A | minus | min Kn | minus | |
| **Thread Plug Gages** | | | | | | | | | | | |
| Go -2B | Y | min Dn | plus | | | | min En | plus | | | Re max |
| No Go -2B | X | min Dn - .05h | minus | | | | max En | minus | | | Rn max |
| Go-pre Plate | Y | min Dn + Ap | plus | | | | min En + Ap | plus | | | Re max |

Symbols used in this table are as follows:

h = Basic depth of thread
Dg = Major dia of gage
Es = Pitch dia of screw
Ds = Major dia of screw
Eg = Pitch dia of gage
A = Allowance of external thread
Ap = Allowance for plating
Kg = Minor dia of thread ring gage
Dn = Major dia of nut
Kn = Minor dia of nut
En = Pitch dia of nut
Rs = Max width of flat of screw minor dia
Rn = Max width of flat of nut minor dia
Ra = Max width of flat of screw major dia
Rc = Min width of flat of screw major dia
Re = Min width of flat of nut minor dia





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TABLE VII. Thread Data (Continued)

| Tolerances for Gage Class | "x" Tolerance | | | "y" Tolerance | | |
|---|---|---|---|---|---|---|
| | 0.05P - 0.1L | 0.1P - 0.2L | .1428P - .2857L | 0.05P - 0.1L | 0.1P - 0.2L | .1428P - .2857L |
| Tolerance in lead | ±.0003 | ±.0003 | ±.0004 | ±.0003 | ±.0003 | ±.0004 |
| Tolerance on half angle of thread | ±15' | ±10' | ±05' | ±20' | ±10' | ±05' |
| Tolerance on major or minor dia | .0005 | .0006 | .0007 | .0005 | .0006 | .0007 |
| Tolerance on pitch dia to and including 1 1/2 inch dia | .0003 | .0003 | .0004 | From .0002 To .0005 | .0002 .0006 | .0002 .0007 |
| Tolerance on pitch dia above 1 1/2 inch dia | .0004 | .0004 | .0005 | From .0002 To .0007 | .0002 .0008 | .0002 .0009 |

FORMULAS FOR CALCULATING THREAD DATA

External Thread (screw)

Major Dia (Max) = Basic thread size - Allowance
(Min) = Max major dia - Tolerance
Pitch Dia (Max) = Max major dia - h
(Min) = Max pitch dia - Tolerance
Minor Dia (Max) = Max major dia - 2h - (0.02P x 2)
(Min) = Max minor dia - Tolerance

Internal Thread (nut)

Minor Dia (Min) = Basic thread size - 1.7648h
(Max) = Min minor dia + Tolerance
Pitch Dia (Min) = Basic thread size - h
(Max) = Min pitch dia + Tolerance
Major Dia (Min) = Basic thread size + (0.02P x 2)
(Max) = Min major dia + Tolerance



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## 3. DEFINITIONS

3.1 Definition and application. The double stub 60° modified thread covered by this standard is in agreement with Handbook H28, insofar as nominal size is concerned. This thread is recommended for electrical connectors and other devices requiring fast coupling action and strong, shallow thread.

## 4. GENERAL REQUIREMENTS

4.1 Not applicable.

## 5. DETAILED REQUIREMENTS

5.1 Form of thread. The double stub 60° modified screw-thread form, as shown in figures 1 and 2, shall be used for all screw-threads covered by this standard.

5.2 Basic thread data. The basic thread data for the double stub 60° modified screw-thread is given in tables VIII, IX and X.

5.3 Thread series. Series of threads are classified and distinguished from each other by (1) the number of threads per inch (pitch) applied to specific diameters and by (2) the relationship of lead to pitch, i.e., lead equal to, double, or triple the pitch.

5.4 Thread classes. Classes of threads are distinguished from each other by the amount of tolerance and allowance specified. Threads specified herein are for the Class 2 double stub 60° modified screw-thread. Figure 2 shows the thread engagement for this class. Class 2A applies to external threads. Class 2B applies to internal threads.

5.5 Thread start. The thread start shall be full width. Full thread depth shall be attained by an increase from root to crest through a maximum arc of 120° (see figure 3).

5.6 Surface roughness. When the requirements of design or application make control of the surface roughness of screw-threads absolutely necessary, the allowable roughness shall be specified in microinches RMS in accordance with MIL-STD-10.

5.7 Gages. All gages shall conform to tables XII, XIII and XIV and the gage tolerances shall conform to applicable requirements of the National Bureau of Standards Handbook H28. The gage design shall be in accordance with Commercial Standard CS8-51 for gage blanks.

5.7.1 Gages for plated threads. Unless otherwise specified, maximum gaging limits of Class 2A products which are plated may be increased by the amount of the product allowance.

5.7.2 Gage wear. Gages shall not be permitted to wear beyond such a point that the product being gaged will be outside the limits specified. Gages shall be reset or replaced when these limits are exceeded.

5.7.3 "No Go" gaging. No go ring or plug gages may engage the product not more than one half turn when the gage shall be restricted to encounter a snug fit for the remainder of the full thread of the product.

5.8 Workmanship. Workmanship shall be consistent with the tolerances specified therein. Threads shall have a smooth finish, free from flaws, abrupt terminations, and other defects which would make them unsuitable for the purpose intended. When the surface roughness is specified, the smoothness of the thread shall be within the limit or limits specified.



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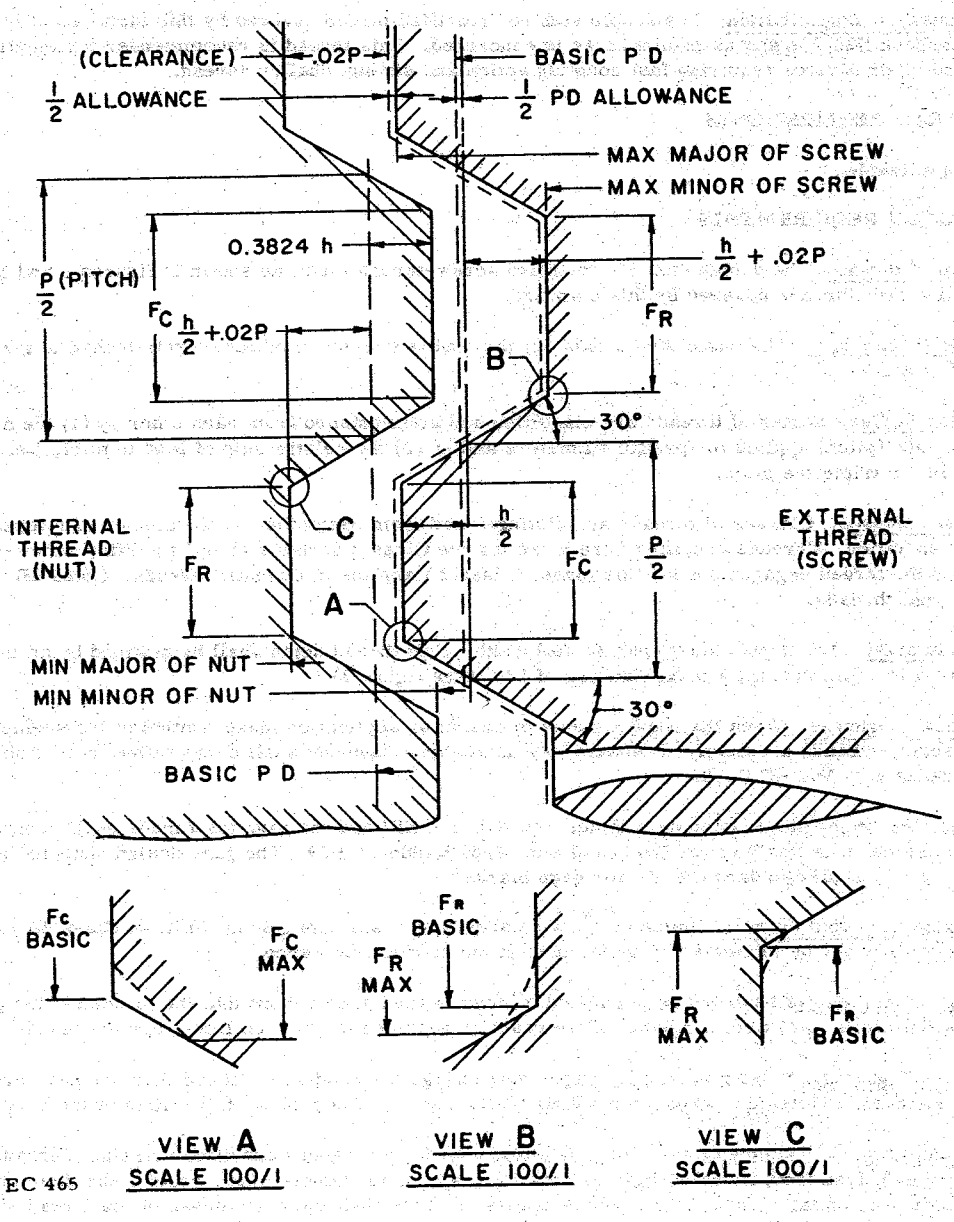
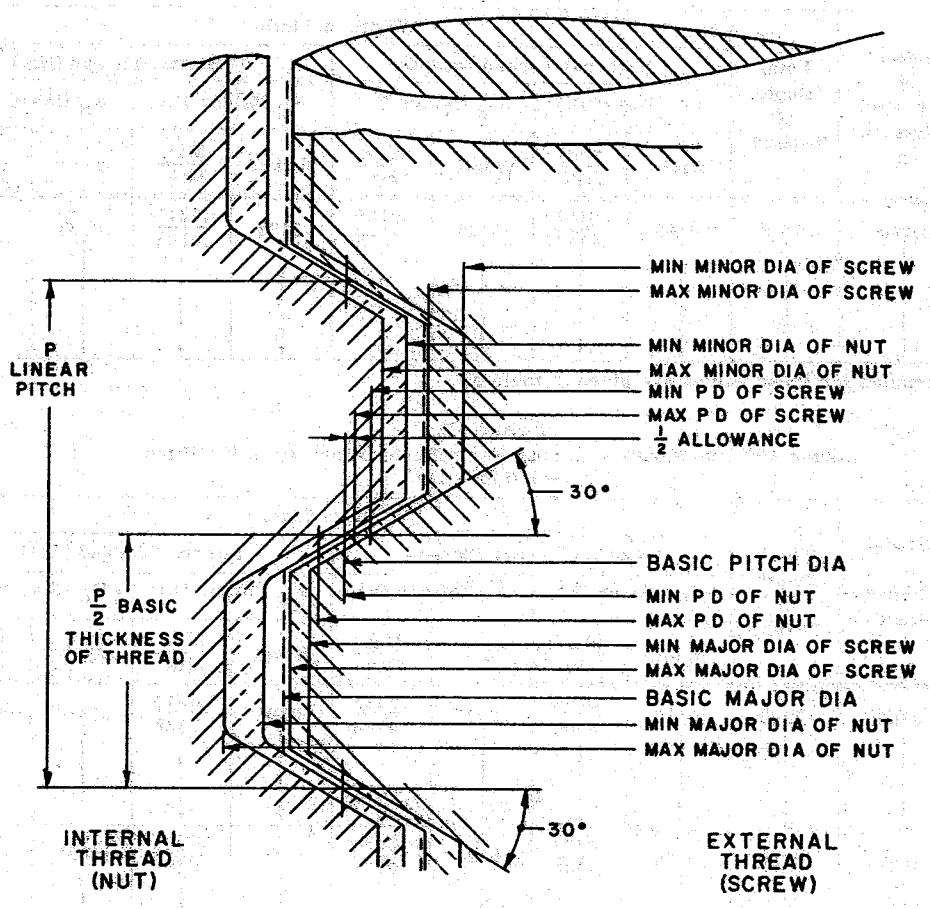



FIGURE 1. Basic dimensions - Class 2A and 2B threads. The dashed line indicates an allowance. Nut and screw threads have been separated for clarity.



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FIGURE 2. Thread engagement for Class 2A and 2B threads. Tolerances, allowances, and crest clearances for Classes 2A and 2B are shown. The dashed line indicates an allowance.



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TABLE VIII.
Double 60° Stub Modified Thread Form Basic Dimensions, .05 Pitch
(See figure 1)

| Thread Size | Depth of Thread (Basic) h | Total Depth of Thread | Width of Flats | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | External Thread (Screw) | | | | Internal Thread (Nut) | | | |
| | | | $F_C$ (Major D.) | | $F_R$ (Minor D.) | | $F_C$ (Minor D.) | | $F_R$ (Major D.) | |
| | | | Basic | Max / Min | Basic | Max / Min | Basic | Max / Min | Basic | Max / Min |
| .2500<br>.3750<br>.5000<br>.5625<br>.6250 | .0180 | .0190 | .0146 | .0181 / .0111 | .0135 | .0169 / .0077 | .0171 | .0217 / .0125 | .0135 | .0181 / .0065 |

Note: Formulas for these values are given in table XI.

TABLE IX.
Double 60° Stub Modified Thread Form Basic Dimensions, 0.1 Pitch
(See figure 1)

| Thread Size | Depth of Thread (Basic) h | Total Depth of Thread | Width of Flats | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | External Thread (Screw) | | | | Internal Thread (Nut) | | | |
| | | | $F_C$ (Major D.) | | $F_R$ (Minor D.) | | $F_C$ (Minor D.) | | $F_R$ (Major D.) | |
| | | | Basic | Max / Min | Basic | Max / Min | Basic | Max / Min | Basic | Max / Min |
| .6875<br>.7500<br>.8750<br>1.0000 | .0260 | .0280 | .0350 | .0396 / .0304 | .0327 | .0373 / .0246 | .0386 | .0443 / .0328 | .0327 | .0385 / .0234 |
| 1.1250<br>1.2500<br>1.3750<br>1.5000<br>1.7500<br>2.0000<br>2.2500<br>2.5000<br>2.6250<br>2.7500<br>3.0000<br>3.2500 | .0340 | .0360 | .0304 | .0373 / .0246 | .0281 | .0338 / .0177 | .0350 | .0419 / .0281 | .0281 | .0350 / .0165 |

Note: Formulas for these values are given in table XI.



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TABLE X.
Double 60° Stub Modified Thread Form Basic Dimensions, .1428 Pitch
(See figure 1)

| Thread Size | Depth of Thread (Basic) h | Total Depth of Thread | Width of Flats | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | External Thread (Screw) | | | | Internal Thread (Nut) | | | |
| | | | $F_C$ (Major D.) | | $F_R$ (Minor D.) | | $F_C$ (Minor D.) | | $F_R$ (Major D.) | |
| | | | Basic | Max / Min | Basic | Max / Min | Basic | Max / Min | Basic | Max / Min |
| 2.0000<br>2.2500<br>2.7500<br>3.0000<br>3.2500 | .0619 | .0647 | .0357 | .0425<br>.0298 | .0324 | .0382<br>.0220 | .0443 | .0512<br>.0373 | .0324 | .0394<br>.0209 |

Note: Formulas for these values are given in table XI.

TABLE XI.
FORMULAS FOR CALCULATING WIDTH OF FLATS

External Thread (screw)

| | |
|---|---|
| $F_C$ (Major Dia) | BASIC = P/2 - [(Max Major D. Screw) - Max P. D. screw)] tan 30° |
| | MAX = P/2 - [(Min Major D. Screw) - (Max P. D. screw)] tan 30° |
| | MIN = P/2 - [(Max Major D. screw) - (Min P. D. screw)] tan 30° |
| $F_R$ (Minor Dia) | BASIC = P/2 - [(Max P. D. screw) - (Max Minor D. screw)] tan 30° |
| | MAX = P/2 - [(Min P. D. screw) - (Max Minor D. screw)] tan 30° |
| | MIN = P/2 - [(Max P. D. screw) - (Min Minor D. screw)] tan 30° |

Internal Thread (nut)

| | |
|---|---|
| $F_C$ (Minor Dia) | BASIC = P/2 - [(Min P. D. nut) - (Min Minor D. nut)] tan 30° |
| | MAX = P/2 - [(Min P. D. nut) - (Max Minor D. nut)] tan 30° |
| | MIN = P/2 - [(Max P. D. nut) - (Min Minor D. nut)] tan 30° |
| $F_R$ (Minor Dia) | BASIC = P/2 - [(Min Major D. nut) - (Min P. D. nut)] tan 30° |
| | MAX = P/2 - [(Min Major D. nut) - (Max P. D. nut)] tan 30° |
| | MIN = P/2 - [(Max Major D. nut) - (Min P. D. nut)] tan 30° |

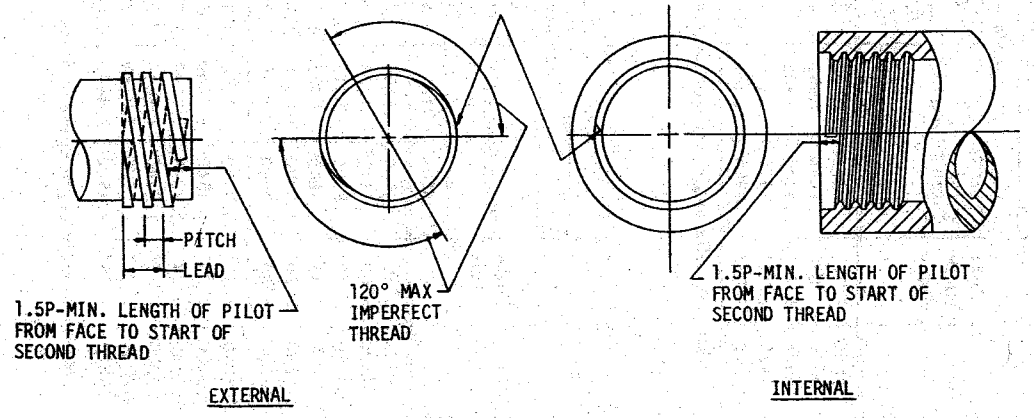



FIGURE 3. Thread start.



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TABLE XII.
Gage Measurements .05 Pitch

| Size | Basic Helix Angle | Setting Plug (External Thread) | | | | | | | | Plug Gage (Internal Thread) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Unplated | | | | After Plating | | | | | | | |
| | | Pitch Diameter | | Measurement Over (3) Wires .02887 Diameter | | Pitch Diameter | | Measurement Over (3) Wires .02887 Diameter | | Pitch Diameter | | Measurement Over (3) Wires .02887 Diameter | |
| | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max |
| .2500 | 7°52' | .2245 | .2305 | .26826 | .27422 | .2245 | .2320 | .26826 | .27572 | .2320 | .2400 | .27572 | .28369 |
| .3750 | 5°07' | .3495 | .3555 | .39299 | .39898 | .3495 | .3570 | .39299 | .40048 | .3570 | .3650 | .40048 | .40847 |
| .5000 | 3°48' | .4745 | .4805 | .51791 | .52391 | .4745 | .4820 | .51791 | .52540 | .4820 | .4900 | .52540 | .53340 |
| .5625 | 3°21' | .5370 | .5430 | .58038 | .58638 | .5370 | .5445 | .58038 | .58788 | .5445 | .5525 | .58788 | .59588 |
| .6250 | 3°01' | .5995 | .6055 | .64287 | .64887 | .5995 | .6070 | .64287 | .65037 | .6070 | .6150 | .65037 | .65837 |

Note: A helix angle correction has been added.

TABLE XIII.
Gage Measurements 0.1 Pitch

| Size | Basic Helix Angle | Setting Plug (External Thread) | | | | | | | | Plug Gage (Internal Thread) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Unplated | | | | After Plating | | | | | | | |
| | | Pitch Diameter | | Measurement Over (3) Wires .05774 Diameter | | Pitch Diameter | | Measurement Over (3) Wires .05774 Diameter | | Pitch Diameter | | Measurement Over (3) Wires .05774 Diameter | |
| | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max |
| .6875 | 5°30' | .6520 | .6600 | .73903 | .74702 | .6520 | .6615 | .73903 | .74852 | .6615 | .6715 | .74852 | .75851 |
| .7500 | 5°02' | .7145 | .7225 | .80146 | .80945 | .7145 | .7240 | .80146 | .81095 | .7240 | .7340 | .81095 | .82094 |
| .8750 | 4°17' | .8395 | .8475 | .92637 | .93436 | .8395 | .8490 | .92637 | .93586 | .8490 | .8590 | .93586 | .94586 |
| 1.0000 | 3°45' | .9645 | .9725 | 1.05131 | 1.05931 | .9645 | .9740 | 1.05131 | 1.06080 | .9740 | .9840 | 1.06080 | 1.07080 |
| 1.1250 | 3°20' | 1.0790 | 1.0890 | 1.16577 | 1.17573 | 1.0790 | 1.1030 | 1.16577 | 1.17776 | 1.0910 | 1.1030 | 1.17776 | 1.18976 |
| 1.2500 | 3°00' | 1.2040 | 1.2140 | 1.29064 | 1.30074 | 1.2040 | 1.2160 | 1.29064 | 1.30274 | 1.2160 | 1.2280 | 1.30274 | 1.31477 |
| 1.3750 | 2°43' | 1.3290 | 1.3390 | 1.41562 | 1.42562 | 1.3290 | 1.3410 | 1.41562 | 1.42762 | 1.3410 | 1.3530 | 1.42762 | 1.43962 |
| 1.5000 | 2°29' | 1.4540 | 1.4640 | 1.54062 | 1.55062 | 1.4540 | 1.4660 | 1.54062 | 1.55262 | 1.4660 | 1.4780 | 1.55262 | 1.56462 |
| 1.7500 | 2°07' | 1.7040 | 1.7140 | 1.79062 | 1.80062 | 1.7040 | 1.7160 | 1.79062 | 1.80262 | 1.7160 | 1.7280 | 1.80262 | 1.81462 |
| 2.0000 | 1°51' | 1.9540 | 1.9640 | 2.04062 | 2.05062 | 1.9540 | 1.9660 | 2.04062 | 2.05262 | 1.9660 | 1.9780 | 2.05262 | 2.06462 |
| 2.2500 | 1°39' | 2.2040 | 2.2140 | 2.29062 | 2.30062 | 2.2040 | 2.2160 | 2.29062 | 2.30262 | 2.2160 | 2.2280 | 2.30262 | 2.31462 |
| 2.5000 | 1°29' | 2.4540 | 2.4640 | 2.54062 | 2.55062 | 2.4540 | 2.4660 | 2.54062 | 2.55262 | 2.4660 | 2.4780 | 2.55262 | 2.56462 |

Note: A helix angle correction has been added for wire dimensions above the dashed line.



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TABLE XIII.
Gage Measurements 0.1 Pitch (Continued)

| Size | Basic Helix Angle | Setting Plug (External Thread) | | | | | | | | Plug Gage (Internal Thread) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Unplated | | | | After Plating | | | | | | | |
| | | Pitch Diameter | | Measurement Over (3) Wires .05774 Diameter | | Pitch Diameter | | Measurement Over (3) Wires .05774 Diameter | | Pitch Diameter | | Measurement Over (3) Wires .05774 Diameter | |
| | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max |
| 2.6250 | 1°24' | 2.5790 | 2.5890 | 2.66562 | 2.67562 | 2.5790 | 2.5910 | 2.66562 | 2.67762 | 2.5910 | 2.6030 | 2.67762 | 2.68962 |
| 2.7500 | 1°21' | 2.7040 | 2.7140 | 2.79062 | 2.88062 | 2.7040 | 2.7160 | 2.79062 | 2.80262 | 2.7160 | 2.7280 | 2.80262 | 2.81462 |
| 3.0000 | 1°14' | 2.9540 | 2.9640 | 3.04062 | 3.05062 | 2.9540 | 2.9660 | 3.04062 | 3.05262 | 2.9660 | 2.9780 | 3.05262 | 3.06462 |
| 3.2500 | 1°8' | 3.2040 | 3.2140 | 3.29062 | 3.30062 | 3.2040 | 3.2160 | 3.29062 | 3.30262 | 3.2160 | 3.2280 | 3.30262 | 3.31462 |

TABLE XIV.
Gage Measurements .1428 Pitch

| Size | Basic Helix Angle | Setting Plug (External Thread) | | | | | | | | Plug Gage (Internal Thread) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Unplated | | | | After Plating | | | | | | | |
| | | Pitch Diameter | | Measurement Over (3) Wires .08248 Diameter | | Pitch Diameter | | Measurement Over (3) Wires .08248 Diameter | | Pitch Diameter | | Measurement Over (3) Wires .08248 Diameter | |
| | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max |
| 2.0000 | 2°42' | 1.9260 | 1.9360 | 2.04972 | 2.05972 | 1.9260 | 1.9380 | 2.04972 | 2.06172 | 1.9380 | 1.9500 | 2.06172 | 2.07372 |
| 2.2500 | 2°23' | 2.1760 | 2.1860 | 2.29972 | 2.30972 | 2.1760 | 2.1880 | 2.29972 | 2.31172 | 2.1880 | 2.2000 | 2.31172 | 2.32372 |
| 2.7500 | 1°56' | 2.6760 | 2.6860 | 2.79972 | 2.80972 | 2.6760 | 2.6880 | 2.79972 | 2.81172 | 2.6880 | 2.7000 | 2.81172 | 2.82372 |
| 3.0000 | 1°47' | 2.9260 | 2.9360 | 3.04972 | 3.05972 | 2.9260 | 2.9380 | 3.04972 | 3.06172 | 2.9380 | 2.9500 | 3.06172 | 3.07372 |
| 3.2500 | 1°38' | 3.1760 | 3.1860 | 3.29972 | 3.30972 | 3.1760 | 3.1880 | 3.29972 | 3.31172 | 3.1880 | 3.2000 | 3.31172 | 3.32372 |





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8 April 1971

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Navy - EC

Review activities:
Army - SC

User activities:
Navy - YD, SH, AS

Preparing activity:
Navy - EC
(Project MISC-0722)

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